CHAPTER 7

TRACE AND DETERMINANT

In this part, we will talk about the trace and determinant of a matrix. These are useful for simplifying calculations involving matrices.

Unlike the previous chapters, we will depart from generic linear transforms and deal exclusively with matrices. In particular, we will use determinants to characterize invertible matrices and find eigenvalues of matrices. Unless otherwise stated, the underlying field $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$.

Trace. Trace is a linear functional on the space of matrices, an element of $\mathcal{L}(\mathbb{F}^{n\times n}, \mathbb{F})$.

DEFINITION 34. *The trace of $A \in \mathbb{F}^{m\times n}$ is* $\operatorname{tr} A = A_{11} + \cdots + A_{nn}$.

We have the following properties of trace, many of which follow immediately from the above definition.

EXERCISE 73. *Show that*

(1) $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ *for all* $A, B \in \mathbb{F}^{m\times n}$.
(2) $\|A\|_F^2 = \operatorname{tr}(A^{\mathrm{H}}A)$ *for all* $A \in \mathbb{F}^{m\times n}$.
(3) $\|u\|^2 = \operatorname{tr}(uu^\top)$ *for all* $u \in \mathbb{R}^n$.
(4) $\|u\|^2 = \operatorname{tr}(uu^{\mathrm{H}})$ *for all* $u \in \mathbb{C}^n$.
(5) $A \sim B \Rightarrow \operatorname{tr} A = \operatorname{tr} B$.

EXERCISE 74. *If A and B are matrix representations of an operator T, show that* $\operatorname{tr} A = \operatorname{tr} B$. *Thus, we can define the trace of T as* $\operatorname{tr} A$, *where* $A = [T]$.

Determinant. We will define determinants using permutations. A permutation on $[n] = \{1, 2, \ldots, n\}$ is a reordering of the elements of $[n]$. We can also view them as bijections on $[n]$.

DEFINITION 35. *A permutation on $[n]$ is a bijection $\sigma : [n] \to [n]$. We will use Π_n to denote the set of permutations on $[n]$.*

EXAMPLE 10. $\Pi_1 = \{(1)\}$ *and* $\Pi_2 = \{(1,2), (2,1)\}$.

EXERCISE 75. *Compute* $|\Pi_n|$.

A permutation can be decomposed into elementary permutations.

DEFINITION 36. *A permutation $\tau \in \Pi_n$ is called a transposition if there exists $p, q \in [n]$, $p \neq q$, such that*

$$\sigma(i) = \begin{cases} q & i = p, \\ p & i = q, \\ i & i \notin \{p, q\}. \end{cases}$$

We will use $\mathcal{T}_n$ to denote the set of transpositions on $[n]$.

EXAMPLE 11. *$(2,1)$ and $(3,2,1)$ are transpositions but $(2,3,1)$ and $(3,1,2)$ are not transpositions.*

EXERCISE 76. *Compute $|\mathcal{T}_n|$ and compare with $|\Pi_n|$.*

The reason we are interested in transpositions is the following result.

EXERCISE 77. *Show that any permutation is a composition of transpositions, i.e.,*

$$\forall\, \sigma \in \Pi_n,\ \exists\, \tau_1, \dots, \tau_k \in \mathcal{T}_n : \quad \sigma = \tau_1 \circ \cdots \circ \tau_k.$$

EXAMPLE 12. *We can write $(3,1,2)$ as the transposition $(1,2,3) \to (1,3,2)$ followed by the transposition $(1,3,2) \to (3,1,2)$.*

There are different ways to express a permutation using transpositions, e.g., applying the same transposition an even number of times does not make any difference. What is invariant is the parity of the number of transpositions in the decomposition.

EXERCISE 78. *Let $\sigma \in \Pi_n$. Suppose there exists $\tau_1, \dots, \tau_k \in \mathcal{T}_n$ and $\rho_1, \dots, \rho_\ell \in \mathcal{T}_n$ such that*

$$\sigma = \tau_1 \circ \cdots \circ \tau_k = \rho_1 \circ \cdots \circ \rho_\ell.$$

Show that both k and ℓ are even or both are odd. Thus, every $\sigma \in \Pi_n$ has even parity (even permutation) or odd parity (odd permutation).

We encode the above structure using a ± 1 signature.

DEFINITION 37. *The sign of a permutation $\sigma \in \Pi_n$ is defined as*

$$\operatorname{sign}(\sigma) = \begin{cases} +1 & \text{if } \sigma \text{ is even}, \\ -1 & \text{if } \sigma \text{ is odd}. \end{cases}$$

EXERCISE 79. *Prove the following results.*

(1) $\operatorname{sign}(\sigma\rho) = \operatorname{sign}(\sigma)\operatorname{sign}(\rho)$ *for all* $\sigma, \rho \in \Pi_n$.
(2) $\Pi_n = \{\sigma\rho : \rho \in \Pi_n\} = \{\rho\sigma : \rho \in \Pi_n\}$ *for all* $\sigma \in \Pi_n$.
(3) $\Pi_n = \{\sigma^{-1} : \sigma \in \Pi_n\}$.

DEFINITION 38. *The determinant of $A = (a_{ij}) \in \mathbb{F}^{n\times n}$ is defined as*

$$\det(A) = \sum_{\sigma \in \Pi_n} \operatorname{sign}(\sigma) \prod_{j=1}^{n} a_{\sigma(j),j}. \tag{11}$$

We collect some useful properties of determinants.

EXERCISE 80. *Let $A, B \in \mathbb{F}^{m\times n}$. Deduce the following properties from definition* (11).

(1) $\det(A) = \sum_{\sigma\in\Pi_n} \operatorname{sign}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}$.
(2) $\det(I_n) = 1$.
(3) $\forall c \in \mathbb{F} :\ \det(cA) = c^n \det(A)$.
(4) $\det(A^\top) = \det(A)$.
(5) $\det(AB) = \det(A)\det(B)$.
(6) $\det(A) = 0 \iff A$ is not invertible.
(7) $A \sim B \Rightarrow \det(A) = \det(B)$.
(8) A has two identical columns (or rows) $\Rightarrow \det(A) = 0$.
(9) A unitary $\Rightarrow |\det(A)| = 1$.

CHAPTER 8

DIAGONALIZATION

In this chapter, we assume that the underlying field $\mathbb{F}$ is the set of complex numbers $\mathbb{C}$. We will use $\mathcal{P}_{\mathbb{C}}$ to denote the space of polynomials with complex coefficients.

Diagonalizability. We wish to understand when an operator can be diagonalized.

DEFINITION 39. *An operator $T \in \mathcal{L}(\mathbb{V})$ is said to be diagonalizable if there exists a basis $v_1, \ldots, v_n$ of $\mathbb{V}$ and $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$ such that*

$$\forall\, 1 \leqslant i \leqslant n : \quad Tv_i = \lambda_i v_i.$$

REMARK 14. *The matrix representation $[T]$ with respect to $v_1, \ldots, v_n$ is diagonal:*

$$A = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}.$$

In other words, a diagonalizable matrix is similar to a diagonal matrix.

Are all operators diagonalizable?

EXERCISE 81. *Let $T \in \mathcal{L}(\mathbb{V})$ be a nonzero operator such that $T^n = 0$ for some $n \geqslant 2$. Show that T cannot be diagonalized. In particular, show that the matrix*

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

cannot be diagonalized as an operator $A \in \mathcal{L}(\mathbb{C}^2)$.

DEFINITION 40. *We say that $\lambda \in \mathbb{C}$ is an eigenvalue of $T \in \mathcal{L}(\mathbb{V})$ if there exists a nonzero $v \in \mathbb{V}$ such that $Tv = \lambda v$. We call v an eigenvector in this case.*

EXAMPLE 13.

(1) *For the zero operator, every nonzero vector is an eigenvector, and 0 is the only eigenvalue*
(2) *For the identity operator, every nonzero vector is an eigenvector, and 1 is the only eigenvalue*
(3) *The eigenvalues of an $(n \times n)$ diagonal matrix are the diagonal elements, and eigenvectors are the columns of the identity matrix I_n.*

EXERCISE 82. *Let T be such that $T^k = 0$ for some $k \geqslant 2$. Find the eigenvalues of T.*

Characteristic polynomial. The following relates eigenvalues with determinants. Recall that a square matrix is not invertible if and only if $\det(A) = 0$.

EXERCISE 83. *Let $T \in \mathcal{L}(\mathbb{V})$ and $\lambda \in \mathbb{C}$. Show that λ is an eigenvalue of T if and only if $T - \lambda \mathrm{Id}$ is not injective. In particular, if $A = [T] \in \mathbb{C}^{n \times n}$, show that $\lambda \in \mathbb{C}$ is an eigenvalue of T if and only if $A - \lambda I_n$ is not invertible, or equivalently, $\det(A - \lambda I_n) = 0$.*

Using the permutation-based formula for determinants, we can easily deduce the following.

EXERCISE 84. *Let $A \in \mathbb{C}^{n\times n}$. Show that $p_A(t) = \det(A - tI_n)$ is a polynomial of the form $p(t) = (-1)^n t^n + a_1 t^{n-1} + \cdots + a_n$.*

The polynomial in question is an important object in the study of eigenvalues.

DEFINITION 41. *Let $A = [T] \in \mathbb{C}^{n\times n}$ be a matrix representation of $T \in \mathcal{L}(\mathbb{V})$. We call $p_A(t) = \det(A - tI_n)$ the characteristic polynomial of T.*

EXERCISE 85. *If A and B are the matrix representations of some operator T, then $p_A = p_B$.*

REMARK 15. *An operator on a vector space over the field $\mathbb{R}$ is not guaranteed to have an eigenvalue. Indeed, for the example*

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

the characteristic polynomial is $p_A(t) = t^2 + 1$, which has no real roots. However, if we switch to $\mathbb{F} = \mathbb{C}$, then A has two eigenvalues.

We have seen that the degree of the characteristic polynomial of an $(n \times n)$ matrix is n. Recall that the fundamental theorem of algebra tells us that any univariate polynomial of degree $p \geqslant 1$ with complex coefficients has exactly p complex roots. Thus, we can conclude that any $(n \times n)$ matrix has n complex eigenvalues including repetitions.

DEFINITION 42. *Let $\lambda \in \mathbb{C}$ be an eigenvalue of T. The algebraic multiplicity of T, denoted as $a(\lambda)$, is the number of times λ appears as a root of the characteristic polynomial.*

In other words, we can write the characteristic polynomial as

$$p_{[T]}(t) = (t - \lambda)^{a(\lambda)} q(t), \qquad q(\lambda) \neq 0.$$

That an operator $T \in \mathcal{L}(\mathbb{V})$ can have at most $n = \dim(\mathbb{V})$ distinct eigenvalues can be deduced from the following observation.

EXERCISE 86. *Let $\lambda_1, \ldots, \lambda_k$ be distinct eigenvalues of T and let $v_1, \ldots, v_k$ be the corresponding eigenvectors. Then $v_1, \ldots, v_k$ are linearly independent.*

We record an important consequence of the above observation – a sufficient condition for an operator to be diagonalizable.

THEOREM 11. *Let $\mathbb{V}$ be finite-dimensional. Then any $T \in \mathcal{L}(\mathbb{V})$ can have at most $\dim \mathbb{V}$ distinct eigenvalues. In particular, T is diagonalizable if it has $\dim \mathbb{V}$ distinct eigenvalues.*

We remark that the last condition is not necessary for diagonalizability. Indeed, the zero and identity transform have just one distinct eigenvalue but are trivially diagonalizable.

Characterization. We will now develop a necessary and sufficient condition for an operator to be diagonalizable. We need the concepts of algebraic and geometric multiplicity in this regard.

DEFINITION 43. *Let $\lambda \in \mathbb{C}$ be an eigenvalue of T. We say that the algebraic multiplicity of λ is k if λ appears k times as a root of the characteristic polynomial. More precisely, we can write the characteristic polynomial as $(t-\lambda)^k q(t)$ for some polynomial q where $q(\lambda) \neq 0$.*

In particular, if $\lambda_1, \ldots, \lambda_k \in \mathbb{C}$ are the distinct eigenvalues of an operator, and $a_1, \ldots, a_k$ are the corresponding algebraic multiplicities, then the characteristic polynomial must be of the form

$$p(t) = (-1)^n (t-\lambda_1)^{a_1} \cdots (t-\lambda_k)^{a_k}.$$

DEFINITION 44. *Let $\lambda \in \mathbb{C}$ be an eigenvalue of T. We call $\mathcal{N}(T - \lambda\,\mathrm{Id})$ the eigenspace of λ and it geometric multiplicity is defined as $g(\lambda) = \mathrm{nullity}\,(T - \lambda\,\mathrm{Id})$.*

A natural question is whether there is a relation between algebraic and geometric multiplicity of an eigenvalue.

EXERCISE 87. *Show that the algebraic multiplicity of an eigenvalue is at least as large as its geometric multiplicity. Moreover, using the example*

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \tag{12}$$

show that the algebraic multiplicity can be larger than the geometric multiplicity.

In fact, we show that the following is true.

EXERCISE 88. *If $\lambda \in \mathbb{C}$ is an eigenvalue of an operator, show that $a(\lambda) \geqslant g(\lambda)$.*

We say that an operator is "defective" if the algebraic multiplicity of one of its eigenvalues is strictly larger than its geometric multiplicity. The following result tells us that these are precisely the operators that cannot be diagonalized.

THEOREM 12. *Let $\lambda_1, \ldots, \lambda_k \in \mathbb{C}$ be the distinct eigenvalues of $T \in \mathcal{L}(\mathbb{V})$. The following are equivalent:*

(1) $g(\lambda_1) + \cdots + g(\lambda_k) = \dim \mathbb{V}$.
(2) $a(\lambda_i) = g(\lambda_i)$ *for* $i = 1, \ldots, k$.
(3) *T is diagonalizable.*

Furthermore, if T is diagonalizable, we have

$$\mathbb{V} = \mathcal{N}(T - \lambda_1 I) \oplus \cdots \oplus \mathcal{N}(T - \lambda_k I).$$

That is, we can be decompose $\mathbb{V}$ into the eigenspaces associated with the distinct eigenvalues.

We can establish Theorem 12 using Exercise 88 and the following result.

EXERCISE 89. *Let $\lambda_1, \ldots, \lambda_k \in \mathbb{C}$ be the distinct eigenvalue of $T \in \mathcal{L}(\mathbb{V})$. Let $u_1, \ldots, u_k$ be such that $u_i \in \mathcal{N}(T - \lambda_i \mathrm{Id})$ for $i = 1, \ldots, k$. Prove that*

$$u_1 + \cdots + u_k = 0 \Rightarrow u_1 = \cdots = u_k = 0.$$

REMARK 16. *As a corollary of Theorem 12, we get the sufficient condition in Theorem 11.*

Schur decomposition. We have seen that not all matrices can be diagonalized. However, every matrix is similar to an upper triangular matrix.

THEOREM 13. *For any $A \in \mathbb{C}^{n\times n}$, we can write $A = U\Lambda U^{\mathrm{H}}$, where $U \in \mathbb{C}^{n\times n}$ is an unitary matrix and an $\Lambda \in \mathbb{C}^{n\times n}$ is an upper triangular matrix.*

This is called the Schur decomposition. It relies on the existence of eigenvalues of a complex matrix. By "upper triangular", we mean $\Lambda_{ij} = 0$ for $i > j$ (elements of Λ below the diagonal are zero). We can interpret this in terms of invariant subspaces, namely, for any $T \in \mathcal{L}(\mathbb{V})$, there exist subspaces

$$\{0\} = \mathbb{V}_0 \subset \mathbb{V}_1 \subset \cdots \subset \mathbb{V}_n = \mathbb{V},$$

where $\dim \mathbb{V}_i = i$ and T is invariant on $\mathbb{V}_i$ for $i = 0, \ldots, n$. By "invariant" we mean that T maps $\mathbb{V}_i$ to itself. In other words, the restriction of T to $\mathbb{V}_i$ is an elemnt of $\mathcal{L}(\mathbb{V}_i)$. It is this invariant structure that gives rise to the upper triangular form of the matrix representation of T.

EXERCISE 90. *Show that the diagonal elements of Λ in the Schur decomposition are exactly the eigenvalues of A.*

EXERCISE 91. *Establish the Schur decomposition for any $A \in \mathbb{C}^{2\times 2}$.*

We will establish the Schur decomposition using induction on the dimension of the matrix. We will also need the following observation for the induction part.

PROPOSITION 6. *Let $T \in \mathcal{L}(\mathbb{V})$ and $\mathbb{W}$ be a subspace of $\mathbb{V}$. Then T is invariant on $\mathbb{W}^{\perp}$ if and only if T^* is invariant on $\mathbb{W}$.*

Cayley-Hamilton theorem. We conclude by discussing this important result. The concept of a matrix polynomial is required to state the result. For any polynomial $h(t) = a_0 + a_1 t + \cdots + a_m t^m$, $a_i \in \mathbb{C}$, and $A \in \mathbb{C}^{n\times n}$, we define the matrix $h(A) \in \mathbb{C}^{n\times n}$ as

$$h(A) = a_0 I_n + a_1 A + \cdots + a_m A^m.$$

THEOREM 14. *Let $A \in \mathbb{C}^{n\times n}$. Then $p_A(A) = 0$.*

EXERCISE 92. *Establish Theorem 14 for a diagonalizable matrix.*

We will use a density argument along with the Schur decomposition to extend the above result to general matrices.

LEMMA 2. *Diagonalizable matrices are dense in $\mathbb{C}^{n\times n}$. That is, for any $A \in \mathbb{C}^{n\times n}$, there exists a sequence of diagonalizable matrices $(A_k)_{k\in\mathbb{N}}$ in $\mathbb{C}^{n\times n}$ such that*

$$\lim_{k\to\infty} \|A_k - A\| = 0,$$

where $\|\cdot\|$ is any matrix norm on $\mathbb{C}^{n\times n}$.

Using Lemma 2 and Exercise 92, we can easily establish Theorem 14. Indeed, the characteristic polynomial $p_A(t)$ is continuous with respect to both A (which determines its coefficients through polynomial functions) and the variable t. Hence, for any $A \in \mathbb{C}^{n\times n}$, considering the sequence $(A_k)_{k\in\mathbb{N}}$ in Lemma 2, we get

$$p_A(A) = \lim_{k\to\infty} p_{A_k}(A_k) = 0.$$